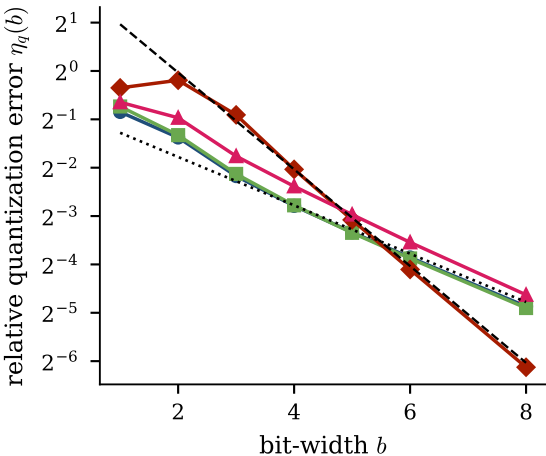
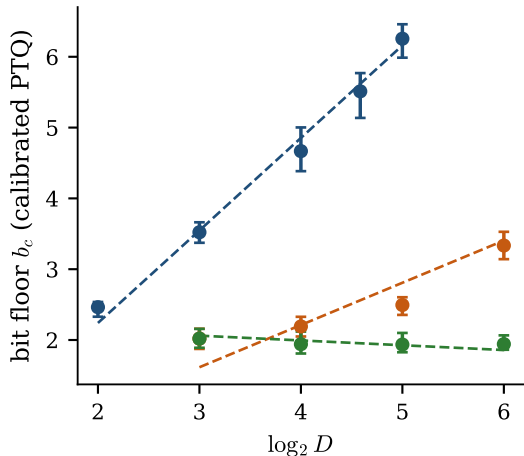


(a) bits-to-noise rate



- MLP, LSQ-style ($\gamma = 0.54$) ▲ ViT, LSQ-style ($\gamma = 0.57$)
- CNN, LSQ-style ($\gamma = 0.55$) theory $\gamma = \frac{1}{2}$
- ◆ CNN, min-max ($\gamma = 1.04$) -.- theory $\gamma = 1$

(b) MLP bit floors (10 seeds)



- - prediction $\alpha_{\text{PTQ}}/\gamma$ ● MLP res. $s = 1/\sqrt{8}$
- MLP plain ● MLP res. $s = 1/\sqrt{D}$